

Prediction of Renewable Energy Generation with Machine Learning: Case Study of the April 2025 Blackout

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Abstract— This work develops and compares three Machine Learning models (Prophet, XGBoost, and LSTM) to predict solar and wind energy generation in the Iberian Peninsula. The instability episodes that culminated in the blackout of 28 April 2025 serve as the case study for this research. The models are trained on hourly generation data from the ENTSO-E Transparency Platform combined with meteorological variables from five representative stations, weighted by installed capacity. The main objective is to assess whether the models detect anomalous deviations from expected generation during critical situations, and to identify which approach is best suited for anomaly detection in renewable generation systems. The results provide recommendations for integrating predictive systems into renewable energy grid monitoring.

Keywords— Machine Learning, renewable energy forecasting, solar photovoltaic, wind energy, LSTM, XGBoost, Prophet, time series, electrical grid, anomaly detection.

1 INTRODUCTION

THE growing integration of renewable energy sources into electrical systems is one of the pillars of the energy transition, but introduces significant technical challenges. In April 2025, Spain experienced a sequence of grid instability episodes that culminated in a widespread blackout on the 28th, preceded by alerts on the 22nd and 24th [6]. Reports from both the Analysis Committee and Universidad Pontificia Comillas attribute the blackout to a combination of factors: insufficient synchronous generation, low inertia due to high renewable penetration, poorly managed frequency oscillations, and weaknesses in the transmission infrastructure [6, 11].

Solar photovoltaic and wind generation depend directly on meteorological conditions, which are variable and difficult to predict with conventional methods [1, 3]. Unlike thermal or nuclear power plants, renewable installations connected through inverters do not contribute rotational inertia to the system, making it harder to stabilise frequency in the event of disturbances [13].

Machine Learning (ML) models can learn complex patterns from historical data and capture non-linear relationships between meteorological variables and generation [7, 14]. In this context, the present work takes a retrospective approach: training three models (Prophet, XGBoost, and LSTM) on 2023–2024 data and checking whether their predictions for April 2025 produce anomalously high errors during the critical episodes, which would evidence external factors that the available meteorological variables cannot capture.

The paper is structured as follows: Section 2 details the objectives, Section 3 presents the theoretical framework, Section 4 describes the methodology, Section 5 reports the results and discussion, and Section 6 concludes the work.

2 OBJECTIVES

The general objective is to develop and compare ML models to predict renewable energy generation and assess their capacity to detect anomalous deviations from expected generation, using the April 2025 episodes as a case study.

The specific objectives are:

1. Collect and preprocess generation data (ENTSO-E) and meteorological variables (Open-Meteo) for the period 2023–2025.
2. Implement and train three models: Prophet, XGBoost, and LSTM.

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3. Evaluate each model using the MAE and RMSE metrics and its anomaly detection capacity.
4. Analyse whether the models generate anomalously high errors during the blackout of the 28th, indicating non-meteorological external factors.
5. Compare the models and identify which is most suitable for anomaly detection in renewable generation monitoring systems.

The scope is limited to the prediction of solar and wind generation at peninsular scale using public data. Out of scope are: the development of a real-time system, the economic analysis of implementation, and the design of energy storage solutions.

3 THEORETICAL FRAMEWORK AND STATE OF THE ART

3.1 Renewable generation prediction with ML

The prediction of renewable energy generation has become a central topic in the context of the energy transition, as these sources carry increasing weight in the electrical system. However, it is a complex task: production depends on factors such as solar irradiance and wind speed, which are highly variable and whose relationship with renewable energy generation is strongly non-linear, making them difficult to predict using conventional methods.

Machine learning models have become a real alternative to traditional approaches: they offer better results in terms of accuracy and adapt better to changing data [7]. This is because they can capture complex relationships between variables that classical statistical models do not detect. Deep learning has further advanced this path, identifying temporal patterns in environments as variable as renewable generation [14].

A key element is the inclusion of meteorological variables as external predictors. Models that use only the time series itself prove insufficient to capture the physical nature of renewable generation. In contrast, including variables such as solar irradiance or wind speed significantly improves the explanatory and predictive capacity of the models [1, 3].

Regarding methodologies, different complementary approaches exist. Prophet is useful for capturing multiple trends and seasonalities with high interpretability; XGBoost stands out for its robustness on tabular data and its capacity to model non-linear relationships; and LSTM networks are especially suited to capturing complex temporal dependencies without the need to manually define lags [12, 2, 5]. Comparing these three models allows an analysis of which approach best suits renewable generation prediction and, at the same time, the detection of anomalous behaviour when real generation diverges from the meteorologically expected prediction.

3.2 Selected models

Prophet [12] is a time series model with an additive architecture that decomposes the series into trend, multiple

seasonality, and holiday effects. Its interpretability makes it especially useful as a baseline.

XGBoost [2] is an *ensemble* algorithm based on decision trees, robust against overfitting and with excellent performance on structured tabular data.

LSTM (*Long Short-Term Memory*) [5] is a recurrent neural network architecture designed to capture long temporal dependencies and complex non-linear relationships between multiple variables.

3.3 The blackout of 28 April 2025

On 28 April 2025, the Iberian Peninsula suffered a widespread electrical blackout, unprecedented in modern times, preceded by instability episodes on 22 and 24 April [6]. The technical report of the Universidad Pontificia Comillas [11] identifies the main causes as low system inertia due to high renewable penetration, uncompensated frequency oscillations, and limitations in the transmission infrastructure. This case is particularly relevant for assessing the capacity of the models to detect anomalous deviations under real critical conditions.

Fig. 1 clearly illustrates the anomaly: on 28 April, cloud cover was below 16% and solar irradiance reached 912 W/m², yet solar generation fell from ~19,000 MW to ~5,000 MW in less than two hours from 10:33 UTC (12:33 local time), the moment of collapse.

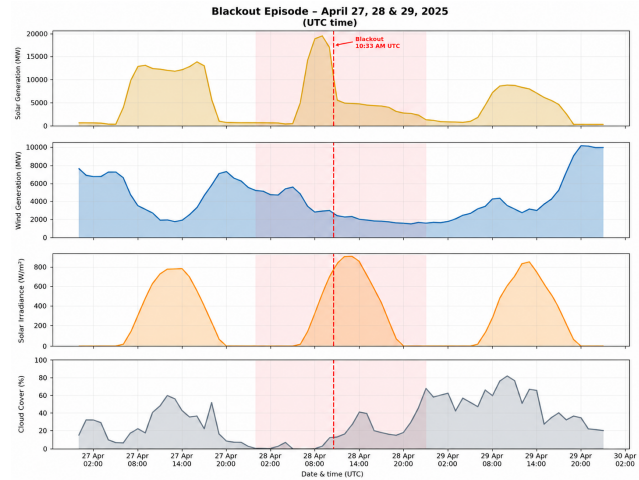


Fig. 1: Blackout episode (27–29 April 2025, UTC time). From top to bottom: solar generation (MW), wind generation (MW), solar irradiance (W/m²), and cloud cover (%). The red line marks 10:33 UTC on 28 April (collapse). The shaded area corresponds to the 28th. Irradiance is at its maximum and cloud cover at its minimum at the moment of collapse, while solar generation drops sharply from ~19,000 MW to ~5,000 MW: a deviation inexplicable by meteorological conditions.

3.4 Research hypotheses

Based on the theoretical framework and the case study described, the following research hypotheses are proposed:

H1. Superiority of machine learning models. The machine learning models (XGBoost and LSTM) will present a lower prediction error (MAE and RMSE) than the additive

Prophet model, thanks to their greater capacity to capture non-linear relationships between meteorological variables and renewable generation.

H2. Increase in errors during anomalous episodes.

Prediction errors will increase significantly during the episodes of 28 April 2025 compared to periods without anomalies, evidencing the presence of external factors not capturable by the available meteorological variables.

H3. Advantage of meteorological models in anomaly detection. Models that predict based exclusively on meteorological variables, such as Prophet, will show greater sensitivity in the detection of anomalous deviations than autoregressive models such as XGBoost.

4 METHODOLOGY

4.1 Data sources

The project combines two public sources. The ENTSO-E Transparency Platform [4] provides hourly electrical generation by technology at peninsular scale, downloaded using the `entsoe-py` library. It should be noted that ENTSO-E data do not separate solar photovoltaic generation from solar thermal; both technologies appear aggregated in a single field. This limitation is maintained throughout the project and is explicitly declared as a methodological constraint.

The originally planned source for generation data was the ESIOS system of Red Eléctrica de España, which offers hourly disaggregated data by technology. The problem is that API access requires a personally requested token, and the validation process was not completed within the project deadline. For this reason, the ENTSO-E Transparency Platform was chosen, which offers equivalent data with open access and without registration.

Open-Meteo [8] is the meteorological reanalysis service selected as an alternative to the AEMET API, which does not offer reliable historical hourly data for the required time range (2023–2025). Open-Meteo is free and requires no registration.

4.2 Geographical representation and weighting

ENTSO-E data are aggregated at peninsular scale. To construct a representative national meteorological series, a weighted average of five stations has been implemented, selected for their representativeness of the main renewable generation zones (Table 1).

TABLE 1: METEOROLOGICAL STATIONS AND JUSTIFICATION.

STATION	JUSTIFICATION
Seville	Leader in peninsular solar generation
Zaragoza	Ebro Valley; solar and wind
Burgos	Wind farms in Castilla y León
A Coruña	Highest wind generation in Spain
Madrid	Central representativeness

The weights are calculated from the actual installed capacity per autonomous community as of 31 December

2023, in accordance with the official REE data [9, 10]:

$$\text{Weight}(z) = \frac{\text{Installed MW}(z)}{\sum_{5 \text{ zones}} \text{Installed MW}} \quad (1)$$

For solar photovoltaic, Seville receives the dominant weight (0.40) due to its high irradiance. For wind, Burgos leads (0.35) due to the concentration of wind farms in Castilla y León. Table 2 shows the final weights.

TABLE 2: WEIGHTING FACTORS BY STATION AND TECHNOLOGY.

STATION	SOLAR WEIGHT	WIND WEIGHT
Seville	0.40	0.15
Zaragoza	0.20	0.25
Burgos	0.15	0.35
A Coruña	0.15	0.20
Madrid	0.10	0.05
Total	1.00	1.00

The weighted national meteorological variable is calculated for each instant t as:

$$V(t) = \sum_{i=1}^5 w_i \cdot x_i(t) \quad (2)$$

where w_i is the weight of station i and $x_i(t)$ the value of the variable at that instant. Solar variables (irradiance, cloud cover) use solar weights; wind variables (wind speed) use wind weights; temperature and precipitation have been weighted uniformly (0.20 per station).

4.3 Meteorological dataset

The consolidated national meteorological dataset contains 21,168 hourly records (1 January 2023 – 31 May 2025), with no missing values, and 9 variables (Table 3). Visual inspection of the series confirms coherent seasonal behaviour: solar irradiance shows maxima in summer and minima in winter; wind speed shows greater variability in autumn and winter.

TABLE 3: VARIABLES OF THE WEIGHTED NATIONAL METEOROLOGICAL DATASET.

VARIABLE	UNIT	RELEVANCE
shortwave_rad.	W/m ²	Global irradiance (solar)
direct_rad.	W/m ²	Direct component
diffuse_rad.	W/m ²	Diffuse component
cloud_cover	%	Cloud cover
wind_100m	m/s	Wind at rotor height
wind_10m	m/s	Wind at 10 m
wind_gusts	m/s	Wind gusts
temperature_2m	°C	Ambient temperature
precipitation	mm	Precipitation

4.4 Preprocessing pipeline

The pipeline follows six steps: (1) **Cleaning** of missing values and point anomalies, with linear temporal interpolation for gaps of up to 3 consecutive hours, and

clip to zero for negative generation values (point artefacts from ENTSO-E); (2) **Temporal synchronisation**: ENTSO-E *timestamps* are in UTC, while Open-Meteo timestamps are in local time (Europe/Madrid), with seasonal clock changes (CET/CEST) that generate ambiguous and non-existent hours. The conversion is done with the `pytz` library, explicitly handling each problematic case; (3) **Integration** into a single *DataFrame* indexed by UTC *timestamp*, via *inner join*; (4) **Derived variables**: hour of the day (0–23), day of the week (0–6), month, day of the year, binary indicators for weekend and national holiday, and *lags* of 1 h, 24 h, and 168 h for the generation variables; (5) **Normalisation** with *MinMaxScaler*, fitted exclusively on the training set and subsequently applied to validation and test to avoid *data leakage*; and (6) **Strict temporal split**: training (2023–2024), validation (Jan.–Mar. 2025), and test (Apr.–May 2025, includes the blackout). Chronological order is preserved throughout to avoid *data leakage*.

The final dataset (`dataset_final_2023-2025.csv`) contains approximately 21,000 hourly records and 25 columns. Correlation analysis confirms that solar irradiance (`shortwave_radiation`) presents a correlation of $r = 0.89$ with solar generation, and wind speed at 100 m (`wind_speed_100m`) of $r = 0.82$ with wind generation, validating the physical coherence of the dataset (Fig. 2).

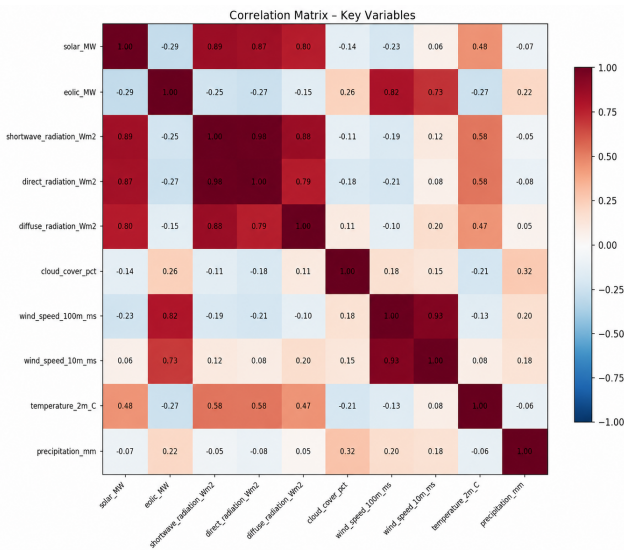


Fig. 2: Pearson correlation matrix between the main variables of the dataset. Notable are the solar–irradiance correlation ($r = 0.89$) and the wind–wind speed at 100 m correlation ($r = 0.82$). The negative correlation between solar and wind ($r = -0.29$) reflects the meteorological complementarity between both sources.

4.5 Machine Learning models

Prophet has been implemented as the reference model for its lower complexity and its capacity to capture multiple seasonalities (annual, weekly, and daily). Meteorological variables have been incorporated as external regressors and Spanish national holidays have been added.

XGBoost is trained on the full tabular dataset, incorporating meteorological variables, derived temporal variables, and autoregressive *lags* of 1 h, 24 h, and 168 h. Hyperpa-

rameter tuning is performed via *early stopping* on the validation set.

LSTM is implemented with a temporal window of 24 hours as input, to capture temporal dependencies without the need for manual *lags*. The architecture includes an LSTM layer of 64 units with *dropout* for regularisation.

4.6 Evaluation metrics

Three error indicators and one anomaly detection criterion are used:

- **MAE** (*Mean Absolute Error*): mean absolute error in MW.
- **RMSE** (*Root Mean Squared Error*): penalises large errors; especially relevant for detecting anomalous peaks.
- **MAPE** (*Mean Absolute Percentage Error*): percentage error, useful for comparing technologies with very different ranges. This metric is excluded for solar generation, since the presence of values very close to zero during night-time hours causes the denominator of the formula to tend to zero, producing numerically unstable values that are not representative of the actual error.
- **Anomaly detection**: analysis of the model residuals on the test set (April–May 2025). A statistical threshold of $\mu + 2\sigma$ calculated from the validation set errors is used. Any test record exceeding this threshold is classified as an anomaly. An abnormally high error under favourable meteorological conditions evidences a non-meteorological external factor.

5 RESULTS AND DISCUSSION

5.1 Prophet

The Prophet model has been trained independently for solar generation and wind generation, incorporating as external regressors the most relevant meteorological variables for each technology: global, direct, and diffuse irradiance, cloud cover, and temperature for the solar model; wind speed at 100 m and 10 m, gusts, temperature, and precipitation for the wind model. Additionally, Spanish national holidays have been added via the `add_country_holidays` function. Seasonality has been configured in multiplicative mode for solar generation, since production is zero during the night, and in additive mode for wind, where this strict daily pattern does not exist. Table 4 shows the evaluation metrics.

TABLE 4: METRICS OF THE PROPHET MODEL FOR SOLAR AND WIND GENERATION.

TECHNOLOGY	SET	MAE (MW)	RMSE (MW)
Solar	Validation	1,346	2,143
Solar	Test	2,060	3,064
Wind	Validation	1,913	2,456
Wind	Test	1,711	2,199

MAPE is excluded as a metric for solar generation for the reasons stated in Section 4.6. For wind generation,

where values do not approach zero, the validation MAPE is 30.79% and the test MAPE is 42.69%.

The results for wind are consistent between validation and test, indicating that the model generalises well. Prophet captures the general trend and seasonal cycles of the wind resource, although it occasionally overestimates intense wind peaks. These peaks are very sudden in nature and difficult to anticipate without knowing future wind conditions.

In contrast, the solar model presents a structural overestimation of midday peaks that is accentuated during the test set (Apr.–May 2025). This discrepancy is due to Prophet extrapolating the growing trend in installed capacity observed during the training period (2023–2024), without being able to account for possible grid operational limitations. In absolute terms, the validation MAE of 1,346 MW represents approximately 7% of the maximum observed solar generation ($\sim 19,000$ MW), an acceptable level for a model in Prophet’s category.

5.1.1 Anomaly detection: 28 April 2025

The residual analysis on the test set reveals the evidence of the blackout unequivocally in the solar component. The anomaly threshold is set at $\mu + 2\sigma$ of the absolute errors on the validation set, as defined in Section 4.6.

28 April 2025 was a meteorologically favourable day: cloud cover below 16% and solar irradiance up to 912 W/m^2 . The model predicts high solar generation, consistent with these conditions. Real generation, however, reached a maximum of around 19,000 MW in the morning and dropped sharply to around 5,000 MW during the central hours of the day, at the moment of the grid collapse. This divergence is inexplicable by the available meteorological variables and constitutes an anomaly of non-meteorological nature: the evidence of the grid collapse (Fig. 3). The model classifies 9 of the 24 hours of 28 April as anomalies (error $> \mu + 2\sigma$).

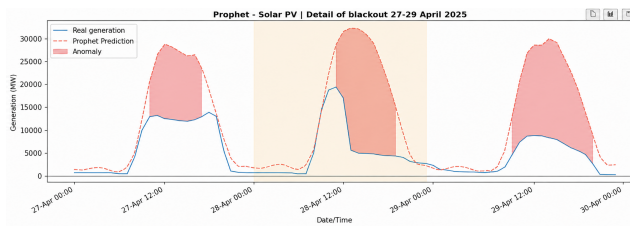


Fig. 3: Detail of the Prophet prediction for solar generation (MW) during 27–29 April 2025. The blue line represents real generation; the dashed red line, the model prediction. The red shaded area indicates hours classified as anomalies (error $> \mu + 2\sigma$). The orange background zone corresponds to the 28th. Solar irradiance is at its maximum and cloudiness at its minimum, but real generation falls to $\sim 5,000$ MW while the model maintains a high prediction.

In contrast, the wind component does not show a clear anomaly on 28 April. The day was anticyclonic, favourable for solar irradiance but adverse for wind, so the low wind generation recorded is partly consistent with the meteorological conditions. The wind model errors remain within the usual range observed during validation, without exceeding the threshold $\mu + 2\sigma$ in a sustained manner (Fig. 4).

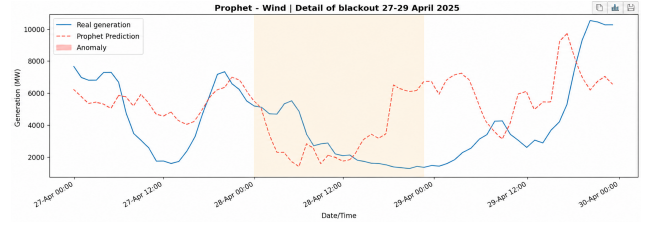


Fig. 4: Detail of the Prophet prediction for wind generation (MW) during 27–29 April 2025. The absence of red areas indicates that no hour on 28 April exceeds the threshold $\mu + 2\sigma$, consistent with the adverse meteorological conditions for wind on that day.

This asymmetry between solar and wind has a clear reading: solar is a strong and unambiguous indicator of the grid collapse, while wind alone does not allow it to be detected.

An additional model limitation should be noted: the holiday component of Prophet partially absorbs the blackout evidence, as the model interprets the generation drop as an effect associated with the proximity of 1 May (national holiday). This confusion between holiday effect and extreme event is an inherent limitation of Prophet’s additive architecture and motivates the implementation of the XGBoost and LSTM models.

5.2 XGBoost

The XGBoost model has been trained independently for solar and wind generation on the full tabular dataset, incorporating meteorological variables, derived temporal variables, and autoregressive *lags* of 1 h, 24 h, and 168 h of the corresponding generation variable. The use of *lags* is justified by the forecasting scenario adopted: *one-step-ahead* prediction with continuous observation, in which the real value at instant $t - 1$ is known when predicting t , as occurs in an operational monitoring system.

Hyperparameter tuning was performed via *early stopping* on the validation set, with a tolerance of 50 iterations without improvement. The final model uses a learning rate of 0.05, maximum depth of 6, and L1 and L2 regularisation. Feature importance was calculated using the *gain* criterion, which measures the average error reduction contributed by each variable. Table 5 shows the evaluation metrics.

TABLE 5: METRICS OF THE XGBOOST MODEL FOR SOLAR AND WIND GENERATION.

TECHNOLOGY	SET	MAE (MW)	RMSE (MW)
Solar	Validation	311	588
Solar	Test	415	730
Wind	Validation	387	530
Wind	Test	378	519

XGBoost outperforms Prophet very significantly in both technologies. For solar, the validation MAE (311 MW) is approximately 4.3 times lower than Prophet’s (1,346 MW), thanks to the model’s capacity to exploit the autocorrelation of the solar series through the *lags*. For wind, the improvement is notable but less pronounced (387 MW vs. 1,913 MW), consistent with the fact that wind generation

exhibits less rigid hour-to-hour autocorrelation than solar. Feature importance analysis (see Figs. 10 and 11 in the appendix) confirms this behaviour: for solar, the 24 h and 168 h *lags* dominate, while for wind, the 1 h *lag* is overwhelmingly dominant.

Notably, XGBoost is, of the two models presented so far, the only one that clearly outperforms the persistence baseline (naïve model: $\hat{y}_t = y_{t-1}$) for solar (311 MW vs. 1,071 MW naïve in validation), while for wind the improvement over the naïve is marginal (387 MW vs. 406 MW naïve), indicating that persistence is already a very competitive predictor for wind generation.

5.2.1 Anomaly detection: 28 April 2025

The residual analysis of the test set reveals a structural limitation of XGBoost for anomaly detection. Applying the $\mu + 2\sigma$ threshold calculated from the validation absolute errors (solar threshold: 1,615 MW; wind threshold: 1,090 MW), the solar model detects only 3 of the 24 hours of 28 April as anomalies (Fig. 5), and the wind model detects none (Fig. 8).

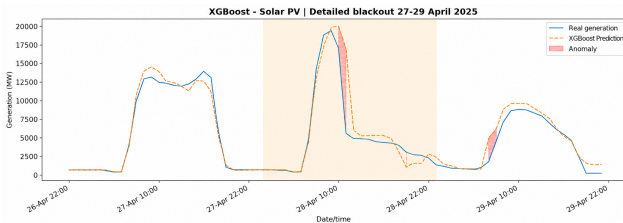


Fig. 5: Detail of the XGBoost prediction for solar generation (MW) during 27–29 April 2025. Unlike Prophet, the model adapts quickly to the generation drop thanks to the autoregressive *lags*, so that only 3 of the 24 hours on 28 April exceed the threshold $\mu + 2\sigma$.

This limitation is a direct consequence of the use of autoregressive *lags*: when real generation drops suddenly due to the blackout, the 1 h *lag* already reflects the new low value, and the model adapts quickly to the new situation instead of maintaining the expected meteorological prediction. This rapid adaptation, which is an advantage under normal conditions, becomes an obstacle to the detection of sudden anomalies.

Unlike Prophet, which relies solely on meteorological variables and therefore maintains a high prediction during the blackout, XGBoost does not capture the divergence between expected and actual generation sufficiently. This fundamental difference motivates the implementation of the LSTM model, which incorporates temporal memory without relying on explicit *lags*.

5.3 LSTM

The LSTM (*Long Short-Term Memory*) model has been trained independently for solar and wind generation. Unlike XGBoost, LSTM does not require manual *lags*: it receives a window of the previous 24 hours as input and automatically learns which temporal dependencies are relevant for the prediction. The forecasting scenario is equivalent to that of XGBoost: *one-step-ahead* with continuous observation.

The implemented architecture consists of an LSTM layer of 64 units, a *dropout* layer with rate 0.2 for regularisation, and two dense output layers (32 and 1 neuron respectively). Data were normalised with *MinMaxScaler*, fitted exclusively on the training set and subsequently applied to validation and test to avoid *data leakage*. Training was performed with *early stopping* on the validation loss, with a tolerance of 10 epochs without improvement and automatic restoration of the best weights (*restore_best_weights=True*).

The management of boundaries between sets is a critical aspect in sequential models: the first 24 validation sequences require the last 24 hours of the training set as context, and analogously for the test set. This concatenation was implemented explicitly to guarantee temporal continuity without introducing future information. Table 6 shows the evaluation metrics.

TABLE 6: METRICS OF THE LSTM MODEL FOR SOLAR AND WIND GENERATION.

TECHNOLOGY	SET	MAE (MW)	RMSE (MW)
Solar	Validation	405	740
Solar	Test	580	970
Wind	Validation	375	518
Wind	Test	350	481

LSTM sits between Prophet and XGBoost in terms of solar accuracy (validation MAE: 405 MW), but outperforms XGBoost for wind in the test set (350 MW vs. 378 MW). For wind, both LSTM and XGBoost improve on the persistence baseline (443 MW naïve at test), while Prophet falls far behind (1,711 MW). For solar, XGBoost is the most accurate model, but LSTM offers a better balance between accuracy and anomaly detection capacity.

5.3.1 Anomaly detection: 28 April 2025

The residual analysis of the test set reveals that LSTM detects 9 of the 24 hours of 28 April as anomalies in the solar component (threshold: 2,136 MW, calculated as $\mu + 2\sigma$ from the validation errors), the same number as Prophet and well above the 3 hours of XGBoost (Fig. 6). For wind, LSTM detects 2 hours, compared to 0 for both XGBoost and Prophet (Fig. 9).

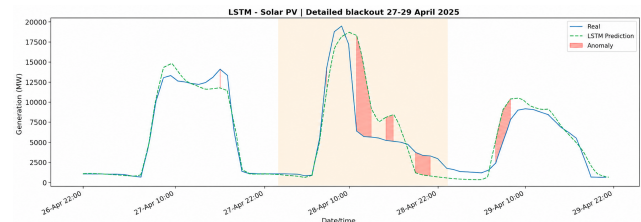


Fig. 6: Detail of the LSTM prediction for solar generation (MW) during 27–29 April 2025. The model detects 9 of the 24 hours on 28 April as anomalies, the same as Prophet, thanks to the absence of explicit *lags* that would allow the model to adapt to the generation drop.

This improvement in anomaly detection relative to XGBoost is a direct consequence of the absence of explicit

lags: the 24-hour window acts as a context memory, but the model maintains its prediction based on the meteorological and seasonal patterns learned during training. When real generation diverges suddenly from these patterns, as occurs during the blackout, the residual is large and the anomaly is clearly detected.

The non-detection for wind is coherent with physical reality: 28 April was an anticyclonic day, adverse for wind, so the low wind generation is partly consistent with the meteorological conditions and does not constitute an unambiguous anomaly (see Fig. 9 in the appendix).

6 CONCLUSIONS

This work has developed and compared three Machine Learning models (Prophet, XGBoost, and LSTM) to predict solar and wind energy generation in the Iberian Peninsula, with the blackout of 28 April 2025 as the central case study. The results allow conclusions to be drawn at two levels: predictive performance and anomaly detection capacity in renewable generation systems.

Predictive performance

In terms of accuracy, XGBoost is the dominant model for solar generation (validation MAE: 311 MW, compared to 405 MW for LSTM and 1,346 MW for Prophet), thanks to the exploitation of autoregressive *lags* in a *one-step-ahead* scenario with continuous observation. For wind generation, LSTM and XGBoost are comparable and both outperform Prophet very significantly (350–378 MW vs. 1,711 MW at test).

The comparison with the persistence baseline ($\hat{y}_t = y_{t-1}$) is especially informative: for solar, XGBoost and LSTM clearly improve on the naïve (311–405 MW vs. 1,071 MW in validation), demonstrating that the models provide real value beyond simple autocorrelation. For wind, in contrast, the improvement over the naïve is marginal for the machine learning models (XGBoost and LSTM: 375–387 MW vs. 406 MW naïve in validation), indicating that persistence is already a very competitive predictor, probably due to the greater temporal smoothness of wind generation compared to solar.

It is important to note that Prophet does not outperform the persistence baseline for solar (1,346 MW vs. 1,071 MW naïve in validation), indicating that its additive architecture is not the most suitable for series with abrupt discontinuities such as the night-time zeros of photovoltaic generation. Table 7 summarises the main metrics for the three models and the baseline.

TABLE 7: COMPARATIVE SUMMARY OF MAE METRICS (MW) FOR ALL MODELS AND THE PERSISTENCE BASELINE. VAL: VALIDATION; TEST: TEST.

MODEL	SOLAR		WIND	
	VAL.	TEST	VAL.	TEST
Prophet	1,346	2,060	1,913	1,711
XGBoost	311	415	387	378
LSTM	405	580	375	350
Naïve	1,071	1,297	406	443

Anomaly detection and case study

The central question of the work is whether the models are capable of detecting the blackout of 28 April 2025. The answer is positive for the solar component, and negative for wind generation, for reasons consistent with physical reality.

28 April was a meteorologically favourable day: cloud cover below 16% and solar irradiance up to 912 W/m². Under these conditions, Prophet and LSTM predicted high solar generation, consistent with the meteorology. Real generation, however, fell from ~19,000 MW to ~5,000 MW in less than two hours from the collapse. This divergence is inexplicable by the available meteorological variables and evidences the action of a non-meteorological external factor.

Prophet and LSTM each detect 9 of the 24 hours on 28 April as solar anomalies, while XGBoost detects only 3. This difference illustrates the *trade-off* between predictive accuracy and sensitivity to anomalies: the autoregressive *lags* of XGBoost cause the model to adapt quickly to the generation drop, reducing the residual and losing the anomaly signature. Prophet and LSTM, in contrast, maintain the expected meteorological prediction and evidence the divergence clearly.

Regarding the proposed hypotheses, the results allow their degree of confirmation to be assessed. *H1* is confirmed: XGBoost and LSTM clearly outperform Prophet in predictive accuracy for both technologies. *H2* is confirmed: the errors on 28 April are much higher than those in the validation period, evidencing the presence of a non-meteorological external factor. *H3* is partially confirmed: Prophet does indeed show greater sensitivity than XGBoost in detecting solar anomalies (9 vs. 3 hours detected), consistent with the hypothesis; but LSTM, despite being a model with temporal memory, achieves the same level of detection as Prophet (9 hours), indicating that the absence of explicit *lags* is the determining factor, rather than the type of architecture.

For wind, no model detects the blackout robustly. 28 April was an anticyclonic day, adverse for wind, so the low wind generation is partly consistent with the meteorology. This result does not represent a limitation of the models but rather an intrinsic characteristic of wind technology in that specific episode.

Recommendation for monitoring systems

The results suggest that, for an anomaly detection system in renewable generation, Prophet and LSTM are more suitable than XGBoost, despite having a higher MAE under normal conditions. The key is that they predict based on meteorological and seasonal patterns, not on immediate autocorrelation, and therefore maintain the meteorologically expected prediction when the grid fails.

The integration of a system of this type into grid monitoring would allow the identification of situations where real generation diverges significantly from the meteorological prediction, alerting operators to possible external factors, such as overloads, infrastructure failures, or extreme events, regardless of the specific cause.

Ethical considerations

Integrating predictive models into electrical grid monitoring is not just a technical problem. If the system generates an alert and the operator acts accordingly, who is responsible if the decision is wrong? Who, if they do not act and the problem worsens? These questions have no easy answer, but they must be kept in mind.

There is an explainability problem that should not be underestimated. Prophet and LSTM detect deviations but do not explain their cause. An operator who receives an alert without context faces a real dilemma: intervene without understanding, or wait and assume the risk. Neither is a comfortable position.

The other risk is excessive confidence. The models were trained on 2023–2024 data. An anomaly of a type they have never seen may go unnoticed, and if the system has been running without errors for some time, the natural tendency is to stop questioning it. That is exactly when it is most dangerous.

These models identify anomalies; they do not make decisions. Interpretation and action must always be the responsibility of a sector expert. An alert system without human supervision does not solve the problem; it displaces it.

Limitations and future work

This work has several limitations. First, ENTSO-E solar generation data combine solar photovoltaic and solar thermal in a single field; disaggregated analysis could improve the interpretation of results. Second, the national meteorological representation is based on five weighted stations, a simplification that may not capture all the spatial heterogeneity of peninsular renewable generation.

The main lines of future work include: (1) the implementation of a real-time alert system based on model residuals, which in a subsequent phase could incorporate parameters related to system inertia and synchronous generation to advance towards an active prevention system; (2) the analysis of other European instability episodes to validate the generalisability of the approach; (3) the incorporation of hybrid models that combine the interpretability of Prophet with the accuracy of XGBoost or LSTM; (4) the extension to other generation technologies to build a more complete grid monitoring system.

Finally, it should be emphasised that none of the implemented models could have prevented the blackout of 28 April on its own: the problem had deep-rooted causes (low system inertia, insufficient synchronous generation, and weak transmission infrastructure) that no renewable generation prediction can substitute. The value of these models is to detect *a posteriori* deviations that are inexplicable by meteorology, not to predict the event itself.

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APPENDIX

A.1 Time series of the national meteorological dataset

Fig. 7 shows the seasonal behaviour of the three main variables of the weighted national meteorological dataset for the period 2023–2025. Solar irradiance shows maxima in summer and minima in winter, wind speed shows greater variability in autumn and winter, and temperature follows the expected annual cycle.

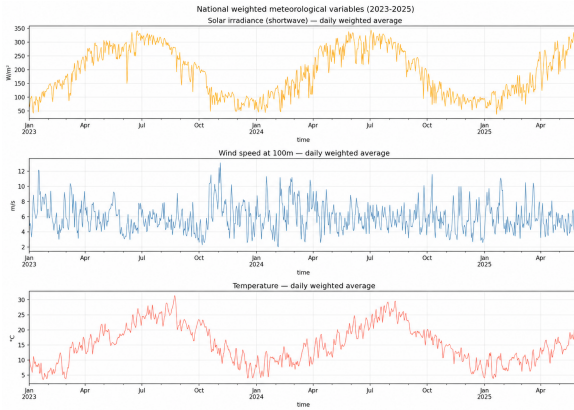


Fig. 7: Weighted national meteorological variables (2023–2025): shortwave solar irradiance (W/m^2), wind speed at 100 m (m/s), and temperature at 2 m ($^{\circ}\text{C}$). Daily averages.

A.2 Detailed results by model

Table 8 shows all MAE and RMSE metrics for the three models and the persistence baseline, in the validation and test sets, for both technologies.

TABLE 8: COMPLETE METRICS (MAE AND RMSE IN MW) FOR ALL MODELS, SETS, AND TECHNOLOGIES.

MODEL	METRIC	SOLAR		WIND	
		VAL.	TEST	VAL.	TEST
Prophet	MAE	1,346	2,060	1,913	1,711
	RMSE	2,143	3,064	2,456	2,199
XGBoost	MAE	311	415	387	378
	RMSE	588	730	530	519
LSTM	MAE	405	580	375	350
	RMSE	740	970	518	481
Naïve	MAE	1,071	1,297	406	443
	RMSE	2,025	2,332	576	622

A.3 Blackout detail: wind component

The following figures show the prediction vs. the real value of wind generation during 27–29 April 2025 for XGBoost and LSTM. For Prophet, see Fig. 4 in the main body of the document. In no case do the detected anomalies coincide with the moment of collapse on 28 April.

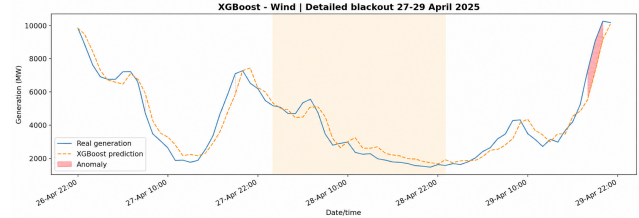


Fig. 8: XGBoost – Wind generation (MW), 27–29 April 2025. The model detects no anomaly on 28 April.

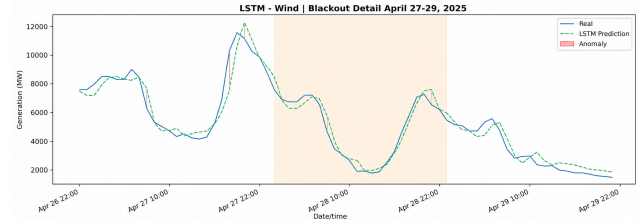


Fig. 9: LSTM – Wind generation (MW), 27–29 April 2025. The model detects 2 anomalous hours, none coinciding with the moment of the collapse.

A.4 Feature importance: XGBoost

Feature importance analysis is specific to XGBoost, as Prophet and LSTM do not have an equivalent direct metric in their architecture.

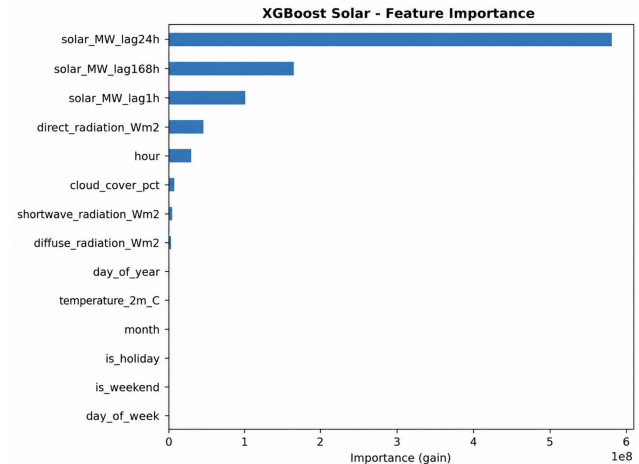


Fig. 10: XGBoost Solar – Feature importance (*gain* criterion). The 24 h and 168 h *lags* dominate, confirming the high autocorrelation of the solar series.

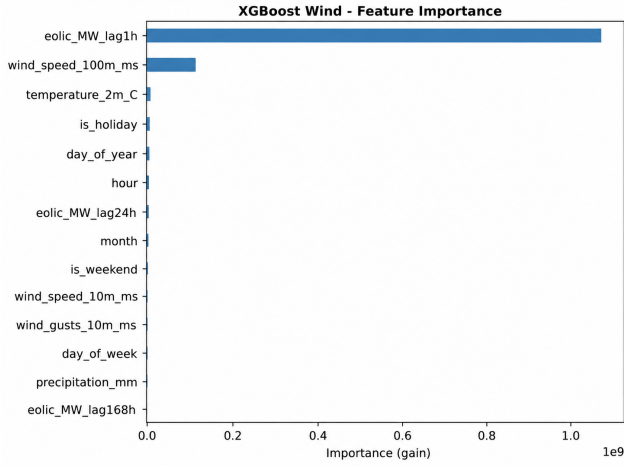


Fig. 11: XGBoost Wind – Feature importance (*gain* criterion). The 1 h *lag* dominates overwhelmingly, explaining why the model adapts quickly to sudden generation changes and does not detect the blackout.

A.5 Key code fragments

The complete code of the training and evaluation notebooks is available at the repository: <https://github.com/arnaupero/bachelor-thesis-renewable-energy-ml>. The most methodologically relevant fragments are reproduced below.

Construction of temporal sequences (LSTM)

The `create_sequences` function transforms tabular data into sequences of length `window_size` for LSTM input. The management of boundaries between sets is performed by concatenating the last `window_size` hours of the previous set to guarantee temporal continuity without introducing future information.

```
def create_sequences(X, y, window_size):
    X_seq, y_seq = [], []
    for i in range(window_size, len(X)):
        X_seq.append(X[i - window_size:i])
        y_seq.append(y[i])
    return np.array(X_seq), np.array(y_seq)

# Validation sequences with training context
X_tv = np.concatenate([X_train_norm, X_val_norm])
y_tv = np.concatenate([y_train, y_val])
X_all, y_all = create_sequences(X_tv, y_tv,
                               WINDOW_SIZE)
X_seq_val = X_all[len(X_seq_train):]
y_seq_val = y_all[len(X_seq_train):]
```

LSTM model architecture

```
def build_model(input_shape, lstm_units=64,
               dropout=0.2):
    model = keras.Sequential([
        layers.Input(shape=input_shape),
        layers.LSTM(lstm_units,
                    return_sequences=False),
        layers.Dropout(dropout),
        layers.Dense(32, activation='relu'),
        layers.Dense(1)
    ])
    model.compile(
        optimizer=keras.optimizers.Adam(
            learning_rate=1e-3),
        loss='mse'
    )
    return model
```

Anomaly detection criterion

The anomaly threshold is calculated as $\mu + 2\sigma$ of the absolute errors on the validation set and applied to the test set.

```
threshold = (residuals_val['residual'].abs().mean()
            + 2 * residuals_val['residual'].abs().std())

residuals_test['anomaly'] = (
    residuals_test['residual'].abs() > threshold
)
```

A.6 Renewable generation time series

Fig. 12 shows the evolution of peninsular solar and wind generation during the period 2023–2025, with the temporal split between the training, validation, and test sets. The growing trend in solar generation is apparent, consistent with the expansion of installed capacity during the period, as is the greater variability of wind generation.

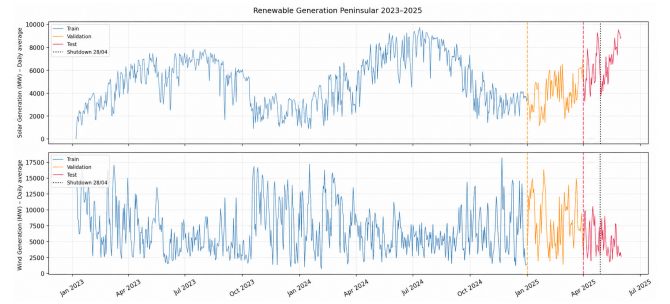


Fig. 12: Peninsular renewable generation (2023–2025): solar (top) and wind (bottom), daily averages in MW. Colours: blue = training, orange = validation, red = test. The dotted line marks the blackout of 28 April 2025.

A.7 Final model hyperparameters

TABLE 9: FINAL HYPERPARAMETERS OF THE THREE MODELS.

MODEL	PARAMETER	VALUE
Prophet	Solar seasonality	Multiplicative
	Wind seasonality	Additive
XGBoost	Learning rate	0.05
	Maximum depth	6
	Regularisation	L1 and L2
	Early stopping	50 iterations
LSTM	LSTM layer units	64
	Dropout	0.2
	Temporal window	24 h
	Early stopping	10 epochs